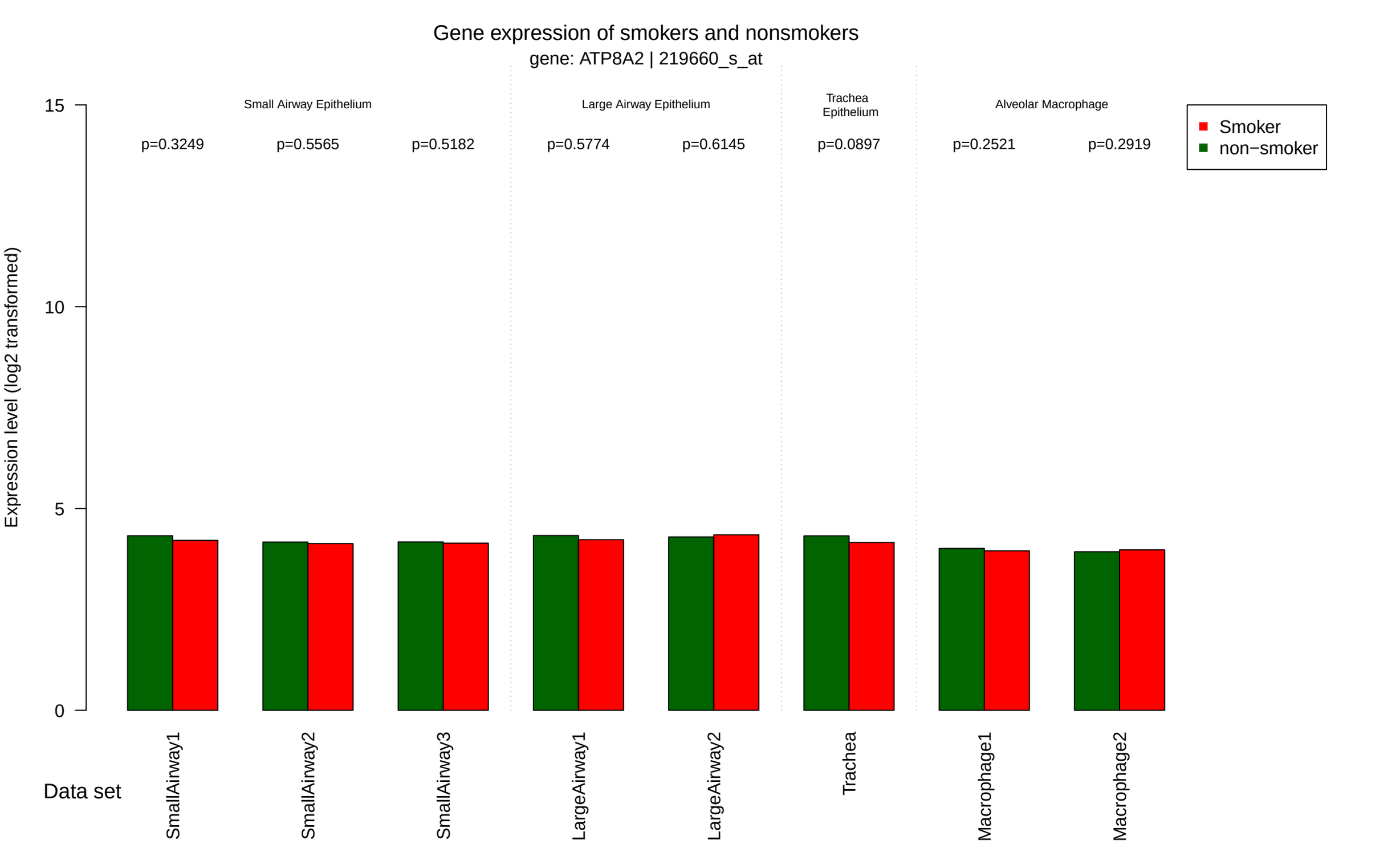
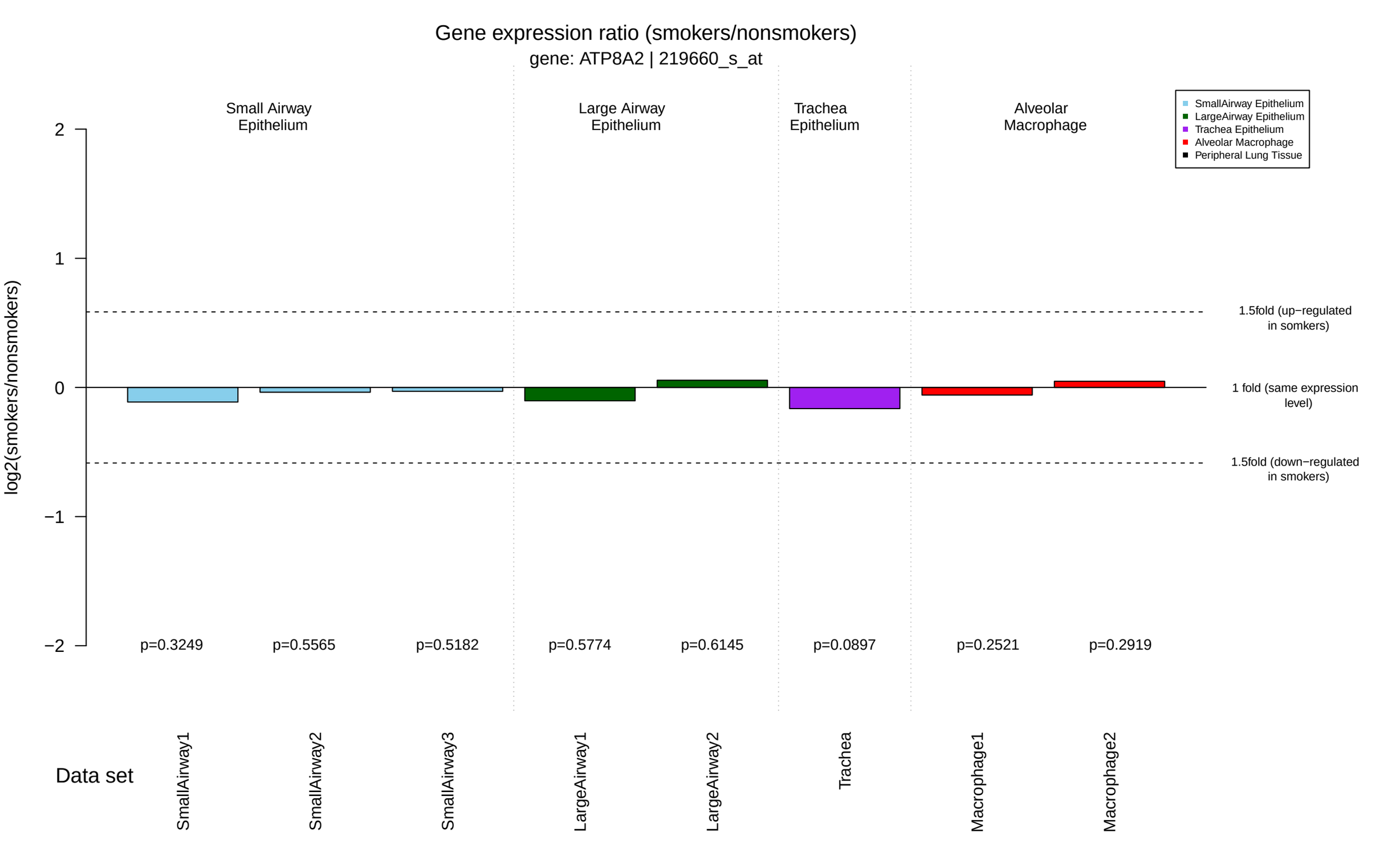
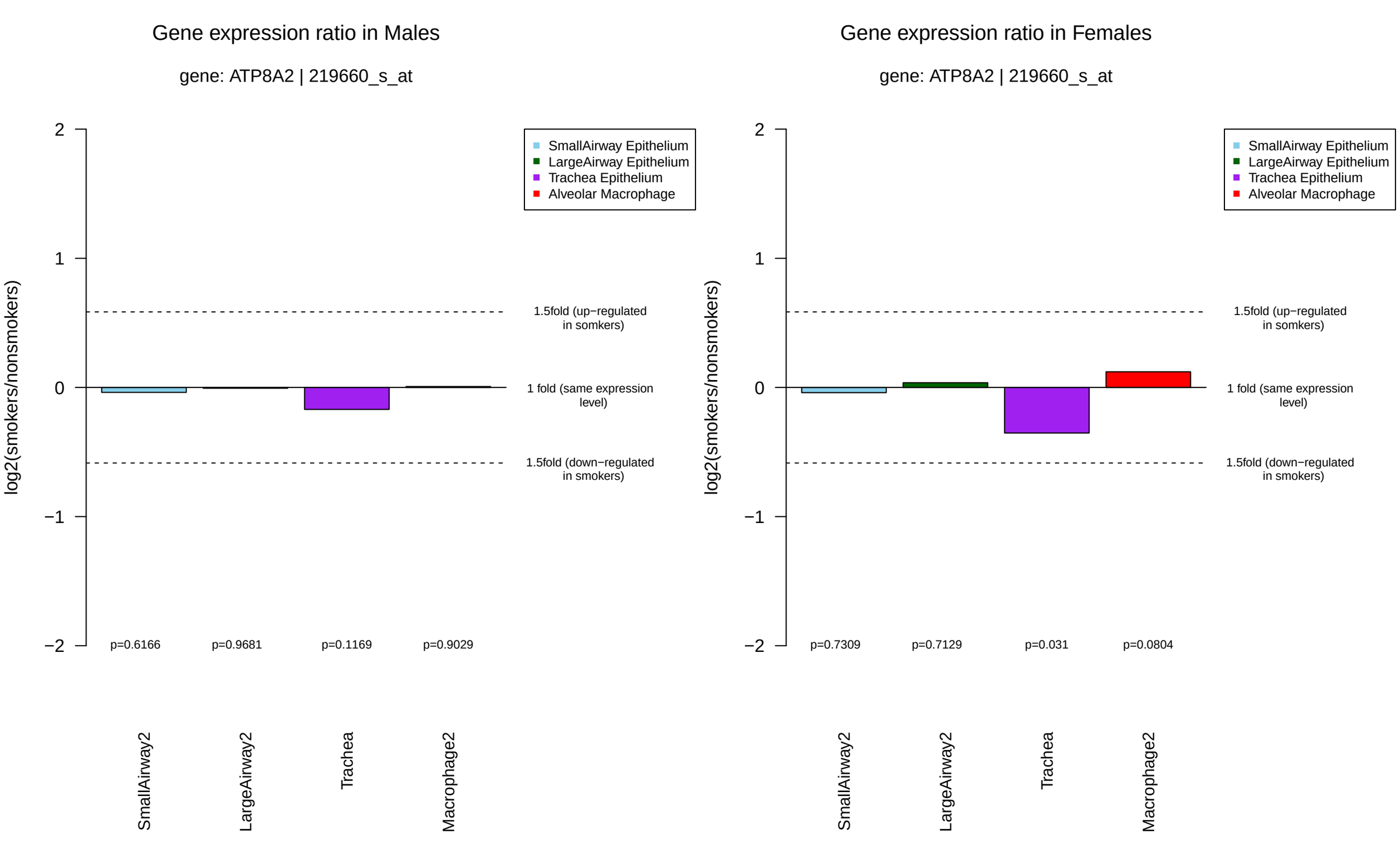
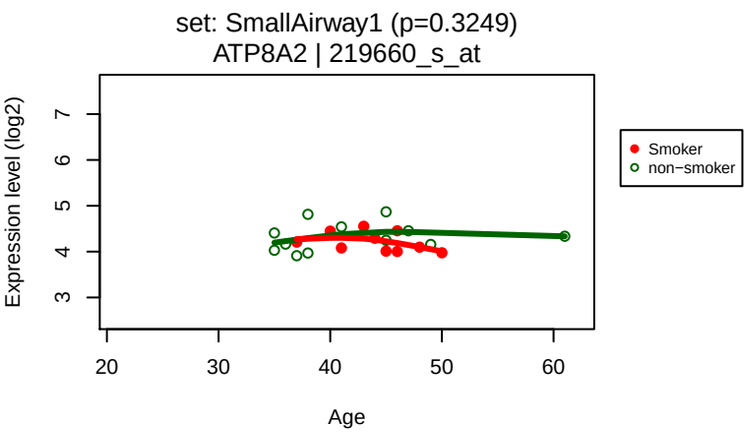
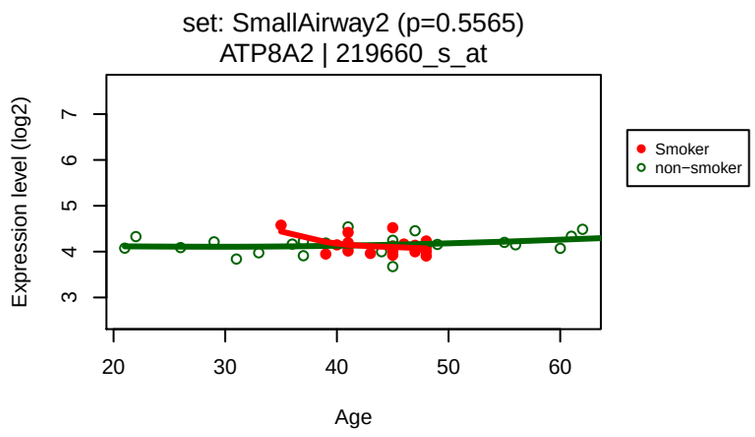
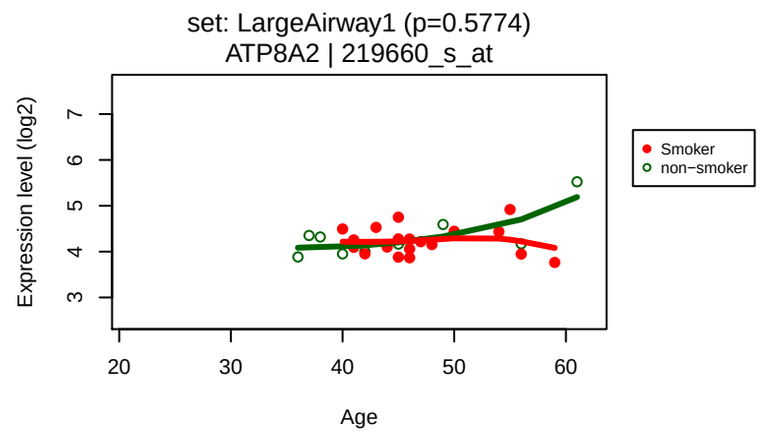

Gene expression of smokers and nonsmokers
gene: ATP8A2 | 219660_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3249
p=0.5565
p=0.5182
p=0.5774
p=0.6145
p=0.0897
p=0.2521
p=0.2919
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

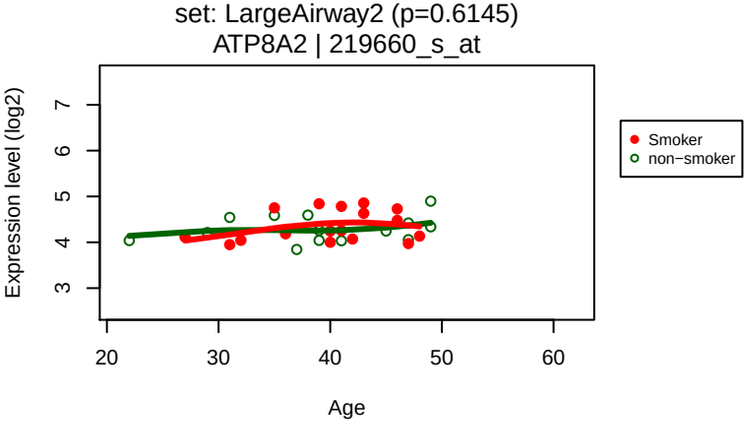

Gene expression ratio (smokers/nonsmokers)
gene: ATP8A2 | 219660_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3249
p=0.5565
p=0.5182
p=0.5774
p=0.6145
p=0.0897
p=0.2521
p=0.2919
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

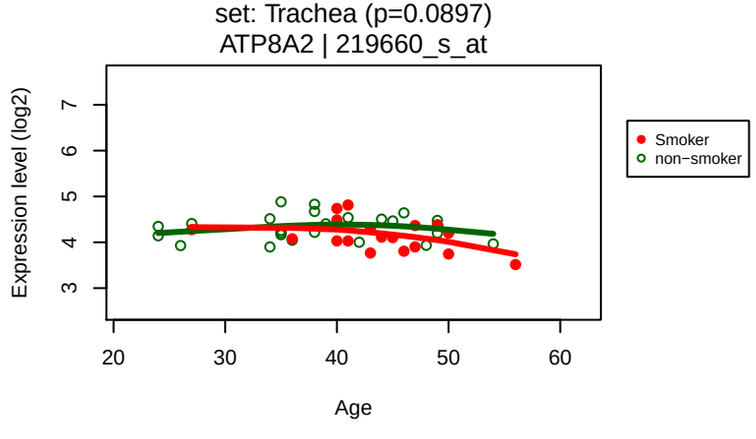

Gene expression ratio in Males
gene: ATP8A2 | 219660_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6166
p=0.9681
p=0.1169
p=0.9029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8A2 | 219660_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7309
p=0.7129
p=0.031
p=0.0804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

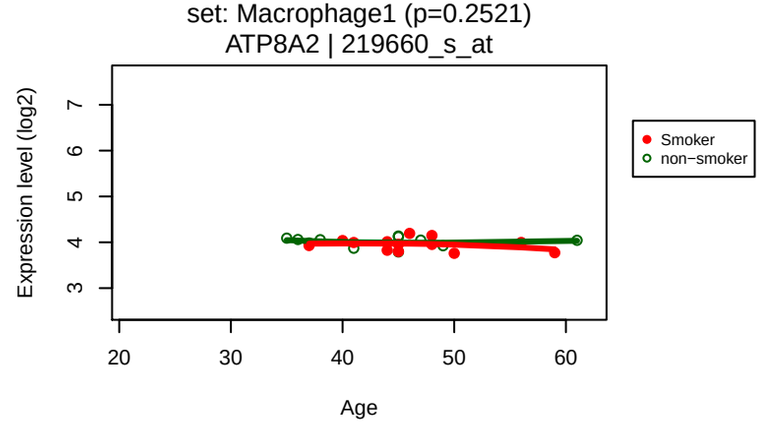

set: SmallAirway1 (p=0.3249)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker

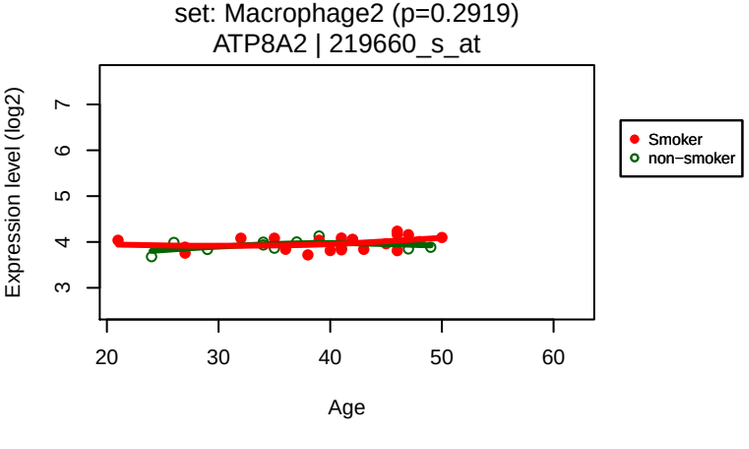

set: SmallAirway2 (p=0.5565)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker

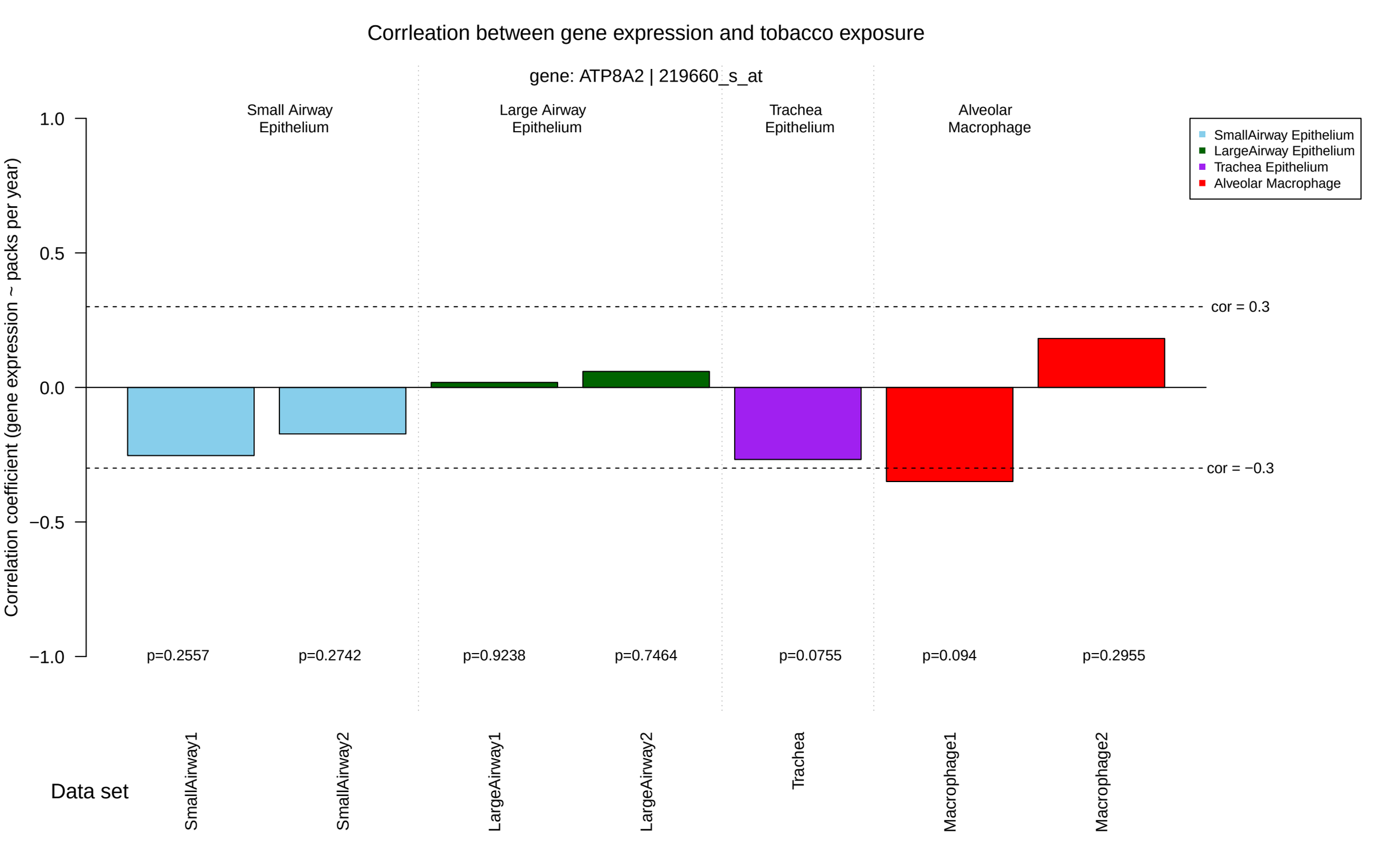

set: LargeAirway1 (p=0.5774)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6145)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0897)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2521)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2919)
ATP8A2 | 219660_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8A2 | 219660_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2557
p=0.2742
p=0.9238
p=0.7464
p=0.0755
p=0.094
p=0.2955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2